

# Data Compression

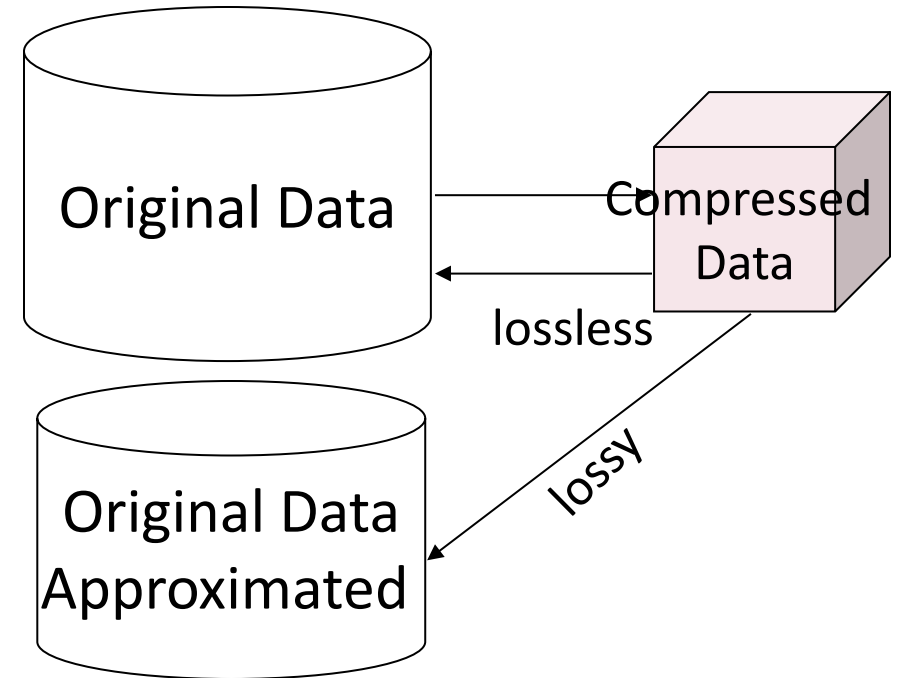
## ❑ String compression

- ❑ There are extensive theories and well-tuned algorithms
- ❑ Typically lossless, but only limited manipulation is possible without expansion

## ❑ Audio/video compression

- ❑ Typically lossy compression, with progressive refinement
- ❑ Sometimes small fragments of signal can be reconstructed without reconstructing the whole
- ❑ Time sequence is not audio
  - ❑ Typically short and vary slowly with time

## ❑ Data reduction and dimensionality reduction may also be considered as forms of data compression



Lossy vs. lossless compression

# Automatic Concept Hierarchy Generation

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- ❑ Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
  - ❑ The attribute with the most distinct values is placed at the lowest level of the hierarchy
  - ❑ Exceptions, e.g., weekday, month, quarter, year

